

Solution Architecture

AI-Assisted Smart Budget Planner

Architecture Flow

Presentation / Client Layer

Static HTML + CSS + JavaScript



Application / API Layer

FastAPI + Uvicorn → Authentication → Planner Routes → Shared Planning Engine



Data / Integration Layer

SQLite (users + plans) | Optional Gemini via HTTPX | Outgoing merchant search links

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects planner inputs and displays results/history.	HTML, CSS, JavaScript
API / Application Layer	Handles routes, validation, authentication and responses.	FastAPI, Uvicorn, Pydantic v2
Authentication	Protects user sessions and supports API authentication.	scrypt, HTTP-only cookies, PyJWT
Core Logic	Allocates whole catalog units without exceeding budget.	Python shared planner / pack()
Database	Stores users and saved plans.	SQLite
AI Integration	Adds optional style/tradeoff commentary only.	Google Gemini + HTTPX
External Search Links	Provides outgoing search URLs; no live price scraping.	Amazon, Flipkart, IKEA, Swiggy, OYO